

Question	Answer	Marks
3(a)(i)	sum of potential and kinetic energies (of molecules/atoms/particles)	B1
	(energy of) molecules/atoms/particles in random motion	B1
3(a)(ii)	(in ideal gas) no intermolecular forces so no potential energy	B1
	internal energy is (solely) kinetic energy (of particles)	B1
	(mean) kinetic energy (of particles) proportional to (thermodynamic) temperature of gas	B1
3(b)	$pV = NkT$	C1
	$6.4 \times 10^6 \times 1.8 \times 10^4 \times 10^{-6} = N \times 1.38 \times 10^{-23} \times 298$	C1
	or	
	$pV = nRT$ and $N = n \times N_A$	(C1)
	$6.4 \times 10^6 \times 1.8 \times 10^4 \times 10^{-6} = n \times 8.31 \times 298$ $n = 46.5$ (mol) $N = 46.5 \times 6.02 \times 10^{23}$	(C1)
	$N = 2.8 \times 10^{25}$	
		A1

9702/41

Cambridge International AS/A Level – Mark Scheme

May/June 2018

PUBLISHED

Question	Answer	Marks
4(c)	a. a. microphones	D1